

Sanjay Mallasamudram Santhanam

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EXPERIENCE

Software Developer

Jul 2021 – Mar 2024

Scry AI

San Jose, CA

- **Tripled backend** throughput, implemented **auto-scaling**, **fault tolerance**, and **disaster recovery** using **Python multiprocessing** library and **Redis** pub-sub
- Cut **parsing errors** from **12% to <1%** by normalizing unstructured data with **NumPy** and **Pandas**
- Reduced **PDF processing time** from **50s to 10s** via **Node.js** refactoring, redesigned **UI** with **HTML**, **CSS** and **Bootstrap**, and secured backend using **AES**, **JWT**, **Express** middleware, **automated sessions**, and **rate limiting**
- Served as **front-end lead**, **mentored 2 junior developers**, boosted **developer productivity** by **40%**, and eliminated code duplication using **AngularJS** modules, a custom **MySQL/MongoDB query builder** and supported **React migration**
- **Wrote 100+ Python unittests**, **fixed critical bugs**, **reduced deployment time** from **1hr to 15min** using **Docker**

Directed Research Intern

May 2020 - Aug 2020

University of Southern California

Los Angeles, CA

- Automated gameplay across **30+** randomly generated environments by building a **Python** Q-learning **AI** for "Spurpunk", a steampunk tower defense game, increasing adaptability.

Research Analyst and Developer

May 2018 - Jun 2018

Teknuance

Chennai, India

- Built an auto-analytics tools using **business intelligence** algorithms and clustered **50K+** data points into high-density groups using clustering algorithms such as **OPTICS** and **DBSCAN**
- Flagged **1200+** anomalies across **6** time series streams using **ARIMA** models in **R**, **Python**, and **Weka**.
- Increased recommendation precision by **18% (top-5 hit rate)** after deploying an **LSTM**-based text recommender and hybrid filtering.

PROJECTS

Signal classification for the Search for Extra Terrestrial Intelligence (SETI) project | Python

- Analyzed spectrogram images of signals from outer space to identify potential indications of intelligent life and proposed an enhanced **InceptionNet** model with **84% accuracy**

Halma AI bot | Artificial Intelligence, Python

- **Developed a Python AI agent** for **HALMA**, a checkers-like strategy board game, using **Alpha-Beta Pruning**, outperforming **700+ students** in competitive gameplay.

Multimedia DBMS | DBMS, SQL, HTML, Javascript, PHP, Python

- Engineered a database management portal using **HTML**, **CSS**, **Javascript**, **PHP**, **Python** to manage student records, implementing **CRUD** functionality for querying **SQL** database.
- Integrated **Python PIL** library to implement **reverse image search** functionality, attaining **85% accuracy**.

EDUCATION

Westcliff University

Irvine, CA

Master of Business Administration, Artificial Intelligence (**Weekend/Professional format**)

Jan. 2025 – Present

University of Southern California

Los Angeles, CA

Master of Science, Computer Science and Engineering

Aug. 2019 – May 2021

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, JavaScript, TypeScript, PHP, Golang, Ruby, R, MATLAB

Web Technologies: HTML5, CSS3, Tailwind CSS, Bootstrap, Angular, React, Next.js, Redux, Vue.js, Node.js Express.js, Flask, Django, FastAPI, jQuery, PHP Laravel, Springboot, .NET, Websockets, Android, Swift

Databases: MongoDB, MySQL, PostgreSQL, Redis, Cassandra, GraphQL, Neo4J, Elasticsearch, Firebase

AI/ Machine Learning: Pandas, NumPy, Keras, PyTorch, TensorFlow, NLTK, Matplotlib, OpenAI, Hugging Face

Testing frameworks: Junit, Selenium, Mockito, Mocha, React testing library, Pytest, Unittest

Tools/ Technologies: Visual Studio Code, Git, Gitlab, Github, Postman, Docker, Kubernetes, AWS, Azure, GCP